

§ Prove Gamma - Poisson Conjugation

$$\text{likelihood: } X \sim \text{Poisson}(N, \lambda), \quad P(X|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\text{prior: } \lambda \sim \text{Gamma}(\alpha, \beta), \quad P(\lambda|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0$$

$$\text{posterior: } P(X|\lambda, \alpha, \beta)$$

$$= \frac{\text{likelihood} \times \text{prior}}{\text{marginal}} = \frac{P(X|\lambda) P(\lambda|\alpha, \beta)}{P(D)}$$

$$= \frac{\frac{\lambda^x e^{-\lambda}}{x!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}}{\int_0^\infty \frac{\theta^x e^{-\theta}}{x!} \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} d\theta}$$

$$= \frac{\lambda^{x+\alpha-1} e^{-(\beta+1)\lambda}}{\int_0^\infty \theta^{x+\alpha-1} e^{-(\beta+1)\theta} d\theta}$$

$$= \frac{(\beta+1)^{x+\alpha}}{\Gamma(x+\alpha)} \lambda^{x+\alpha-1} e^{-(\beta+1)\lambda}$$

$$\sim \text{Gamma}(x+\alpha, \beta+1)$$

By Gamma( $x+\alpha, \beta+1$ )

$$\int_0^\infty \frac{(\beta+1)^{x+\alpha}}{\Gamma(x+\alpha)} \theta^{x+\alpha-1} e^{-(\beta+1)\theta} d\theta = 1$$

$$\Rightarrow \frac{(\beta+1)^{x+\alpha}}{\Gamma(x+\alpha)} \int_0^\infty \theta^{x+\alpha-1} e^{-(\beta+1)\theta} d\theta = 1$$

$$\Rightarrow \int_0^\infty \theta^{x+\alpha-1} e^{-(\beta+1)\theta} d\theta = \frac{\Gamma(x+\alpha)}{(\beta+1)^{x+\alpha}}$$

### § Posterior mean and variance with Gaussian prior

prior:  $w \sim N(\mu_0, \Lambda_0^{-1}) \Rightarrow$  Multivariate Gaussian

Assume  $y \sim N(w^T \Phi, a^{-1}) \Rightarrow$  Univariate Gaussian

By Bayes' rule:

$$P(w|D) \propto P(D|w) \cdot P(w)$$

$$\propto P(y|w^T \Phi, a^{-1}) P(w|\mu_0, \Lambda_0^{-1})$$

$$= e^{-\frac{a}{2}(y - w^T \Phi)^2} \cdot e^{-\frac{1}{2}(w - \mu_0)^T \Lambda_0 (w - \mu_0)}$$

$$= e^{-\frac{a}{2}(y^2 - 2w^T \Phi y + w^T \Phi \Phi^T w) - \frac{1}{2}(w^T \Lambda_0 w - 2w^T \Lambda_0 \mu_0 + \mu_0^T \Lambda_0 \mu_0)}$$

$$= e^{-\frac{1}{2}(ay^2 - 2w^T \Phi y a + a w^T \Phi \Phi^T w + w^T \Lambda_0 w - 2w^T \Lambda_0 \mu_0 + \mu_0^T \Lambda_0 \mu_0)}$$



$w$ :  $k \times 1$  matrix

$\Phi$ :  $k \times 1$  matrix

$\mu_0$ :  $k \times 1$  matrix

$\Lambda_0$ :  $k \times k$  matrix

$y$ : scalar

$a$ : scalar

• Completing the square

将 ② 项 改 成  $(x - \mu')^T \Lambda' (x - \mu')$  的 形 式

$$(x - \mu')^T \Lambda' (x - \mu') = x^T \Lambda' x - 2x^T \Lambda' \mu' + \mu'^T \Lambda' \mu'$$

$$\textcircled{2} = w^T a \Phi \Phi^T w + w^T \Lambda_0 w$$

$$= w^T (a \Phi \Phi^T + \Lambda_0) w, \text{ let } a \Phi \Phi^T + \Lambda_0 = C$$

$$= w^T C w$$

$$\text{Let } w^T C w = x^T \Lambda' x \Rightarrow x = w, \Lambda' = C$$

$$\begin{aligned} \textcircled{3} &= -2 \omega^T \bar{\Phi} y a - 2 \omega^T \lambda_0 \mu_0 \\ &= -2 \omega^T (\bar{\Phi} y a + \lambda_0 \mu_0) \end{aligned}$$

$$\text{Let } -2 \omega^T (\bar{\Phi} y a + \lambda_0 \mu_0) = -2 \chi^T \Lambda' \mu'$$

$$\Rightarrow \bar{\Phi} y a + \lambda_0 \mu_0 = C \mu'$$

$$\Rightarrow \mu' = C^{-1} (\bar{\Phi} y a + \lambda_0 \mu_0)$$

$$\begin{aligned} \textcircled{1} &= \omega^T C \omega - 2 \omega^T C \mu' + \mu'^T C \mu' - \mu'^T C \mu' + ay^2 + \mu_0^T \lambda_0 \mu_0 \\ &= (\omega - \mu')^T C (\omega - \mu') - \mu'^T C \mu' + ay^2 + \mu_0^T \lambda_0 \mu_0 \end{aligned}$$

So the formula become:

$$e^{-\frac{1}{2} [(\omega - \mu')^T C (\omega - \mu') - \mu'^T C \mu' + ay^2 + \mu_0^T \lambda_0 \mu_0]}$$

$$= e^{-\frac{1}{2} (\omega - \mu')^T C (\omega - \mu')} e^{-\frac{1}{2} (-\mu'^T C \mu' + ay^2 + \mu_0^T \lambda_0 \mu_0)}$$

$$\sim N(\mu', C^{-1})$$

$$\mu' = C^{-1} (\bar{\Phi} y a + \lambda_0 \mu_0)$$

$$C^{-1} = (\lambda_0 + a \bar{\Phi} \bar{\Phi}^T)^{-1}$$

By Sherman-Morrison Formula

$$= \left( \lambda_0^{-1} - \frac{\lambda_0^{-1} a \bar{\Phi} \bar{\Phi}^T \lambda_0^{-1}}{1 + a \bar{\Phi}^T \lambda_0^{-1} \bar{\Phi}} \right)$$

◦ Sherman - Morrison Formula

$$C^{-1} = \Lambda_0^{-1} - \frac{\Lambda_0^{-1} a \Phi \Phi^T \Lambda_0^{-1}}{1 + a \Phi^T \Lambda_0^{-1} \Phi}$$

$$( \Lambda_0 + a \Phi \Phi^T ) ( \Lambda_0^{-1} - \frac{\Lambda_0^{-1} a \Phi \Phi^T \Lambda_0^{-1}}{1 + a \Phi^T \Lambda_0^{-1} \Phi} )$$

$$= I + a \Phi \Phi^T \Lambda_0^{-1} - \frac{a \Phi \Phi^T \Lambda_0^{-1} a \Phi \Phi^T \Lambda_0^{-1}}{1 + a \Phi^T \Lambda_0^{-1} \Phi} - \frac{a \Phi \Phi^T \Lambda_0^{-1}}{1 + a \Phi^T \Lambda_0^{-1} \Phi}$$

$$= I + a \Phi \Phi^T \Lambda_0^{-1} - \frac{a \Phi ( \cancel{\Phi^T \Lambda_0^{-1} a \Phi} + 1 ) \bar{\Phi}^T \Lambda_0^{-1}}{1 + \cancel{a \Phi^T \Lambda_0^{-1} \Phi}}$$

$$= I + a \Phi \Phi^T \Lambda_0^{-1} - a \Phi \Phi^T \Lambda_0^{-1} = I$$

◦ For the next problem:

prior:  $w \sim N(\mu_0, \Lambda_0^{-1})$

design matrix:  $\begin{bmatrix} x^* \\ x^* \\ \vdots \\ x^{n+1} \end{bmatrix}$

likelihood:  $y \sim N(w^T \Phi, a)$

posterior:  $w \sim N(\mu', \Lambda'^{-1})$

First prior:  $w \sim N(0, b^{-1} I)$

$$\Lambda'^{-1} = (\Lambda_0 + a^{-1} \Phi \Phi^T)^{-1}$$

$$\mu' = \Lambda'^{-1} (\Phi y a^{-1} + \Lambda_0 \mu_0)$$